

Apply topology optimization (SIMP) to shoulder prosthesis components to minimize weight while preserving stiffness and structural safety without active control elements

Executive Technology Brief

Domain: mechanical_structural

Technology Maturity: ENGINEERING_DEFINITION

Transfer Posture: Sponsored Validation

Problem: Documented recurring problem 'appropriate_device_problem_term_code_not' for shoulder prosthesis: 23 MAUDE reports across 3 distinct devices (20200506 to 20211104); root-cause evidence from 2 recall records and 10 event narratives (custody attached in the campaign report). MAUDE limitations: incidence UNKNOWN, causation UNVERIFIED (Art. XXI.5). Existing solutions fail because: Why existing solutions fail, per retrieved evidence: 2 recall root-cause descriptions (FDA enforcement records, custody attached); 10 MAUDE event narratives from this mechanism cluster (mdr_text, custody attached). Statements are the providers' own words, cust Unresolved gap: Across the 3 relevant retrieved attempts (a bounded, non-exhaustive set), none measurably addresses: appropriate, available, conditions, devices, events, failure, function, injury, intervention, introducing, maintain, matching, modes, must, producing, recurrin

Mechanism: Passive structural optimization with topology optimization

Architecture: PROPOSED architecture (no physical system exists): load path / structural backbone; actuation or compliance element; guidance / bearing interfaces; fatigue-critical features register — realizing: Apply topology optimization (SIMP) to shoulder prosthesis components to minimize weight while preserving stiffness and structural safety without active control elements

Governing Model: Mechanics / structures governing relations proposed for this invention; SYMBOLIC ONLY until inputs are sourced

Key Equations:

MEC-002: Section bending stress. Expression: $\sigma = M * c / I$. Applicability to this invention: CONDITIONAL; holds under: linear-elastic material; small deflection; prismatic section. Assumptions: stress concentrators multiply local stress above the nominal value – geometry factors must be applied from the actual design. Engaged variables: NOT ESTABLISHED.

MEC-003: Hooke's law (linear elasticity). Expression: $\sigma = E * \epsilon$. Applicability to this invention: CONDITIONAL; holds under: below proportional limit; isothermal. Assumptions: superelastic members (e.g. nitinol) do not follow linear elasticity through transformation – use their measured plateau relations instead. Engaged variables: NOT ESTABLISHED.

MEC-001: Euler buckling load. Expression: $P_{cr} = \pi^2 * E * I / (K * L)^2$. Applicability to this invention: CONDITIONAL; holds under: elastic, slender column, small deflection. Assumptions: K for catheter constraint conditions is MODEL_DERIVED. Engaged variables: NOT ESTABLISHED.

Kill Condition: killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.44573539382492045, EIG-per-cost 0.44573539382492045; the falsification test carries the pre-registered pass/fail before any transfer)

Current State: ENGINEERING_DEFINITION. No physical prototype. No clinical validation.

Disclosure: This dossier distinguishes established evidence from proposals and unresolved questions.